

**PHYSICS – BOARD QUESTION PAPER (JUNE 2025)****Time: 3 Hours****Max. Marks: 70****General Instructions:**

1. Section A: Q1 contains Ten MCQs (1 mark each). Q2 contains Eight very short answer questions (1 mark each).
2. Section B: Q3 to Q14 contain Twelve short answer questions (2 marks each). Attempt any EIGHT.
3. Section C: Q15 to Q26 contain Twelve questions (3 marks each). Attempt any EIGHT.
4. Section D: Q27 to Q31 contain Five long answer questions (4 marks each). Attempt any THREE.
5. Use of log table is allowed. Calculator is not allowed.
6. Figures to the right indicate full marks.
7. For MCQs, only the first attempt will be considered.

**Physical Constants:**

- (i)  $h = 6.63 \times 10^{-34} \text{ Js}$
- (ii)  $c = 3 \times 10^8 \text{ m/s}$
- (iii)  $\pi = 3.142$
- (iv)  $g = 9.8 \text{ m/s}^2$
- (v)  $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$
- (vi)  $H = 4\pi \times 10^{-7} \text{ Wb/A-m}$
- (vii)  $\sigma = 5.67 \times 10^{-8} \text{ J/m}^2\text{sK}^4$
- (viii)  $R_H = 1.097 \times 10^7 \text{ m}^{-1}$
- (ix) Mass of electron =  $9.11 \times 10^{-31} \text{ kg}$
- (x) Mass of proton =  $1.672 \times 10^{-27} \text{ kg}$

**SECTION – A****Q1. Select and write the correct answer: (10M)**

- i. Photoelectric effect is possible for visible radiation for which metal?  
(A) Zinc (B) Cadmium (C) Magnesium (D) Sodium
- ii. Gyromagnetic ratio ( $e/m$ ) of an electron is:  
(A)  $e/2m$  (B)  $e/m$  (C)  $2e/m$  (D)  $e/2m_e$



iii. In a cyclotron, final kinetic energy of a particle at radius  $R_{\text{exit}}$  is:

- (A)  $K.E. = eB^2R_{\text{exit}}^2$
- (B)  $K.E. = (e^2R_{\text{exit}}^2)/m$
- (C)  $K.E. = (e^2B^2R_{\text{exit}}^2)/(2m)$
- (D)  $K.E. = (e^2R_{\text{exit}}^2)/(2m)$

iv. The equation  $y = a \sin(kx - \omega t)$  represents:

- (A) Progressive wave along +x direction
- (B) SHM
- (C) Standing wave
- (D) Progressive wave along -x direction

v. Principle of perpendicular axes is applicable for:

- (A) Spherical bodies
- (B) Cylindrical bodies
- (C) Conical bodies
- (D) Laminar bodies

vi. Two capillary tubes of radii 0.3 cm and 0.6 cm are dipped in same liquid. Ratio of heights  $h_1 : h_2$  is:

- (A) 1:2
- (B) 2:1
- (C) 1:4
- (D) 4:1

vii. According to Kirchhoff's Voltage Law, the algebraic sum of emf around a closed loop equals:

- (A) Momentum
- (B) Mass
- (C) Charge
- (D) Zero

viii. The intensity of unpolarized light after passing through a polarizer becomes:

- (A)  $I/2$
- (B)  $2I$
- (C)  $I^2$
- (D)  $I/4$

ix. A particle performing linear SHM has acceleration equation  $a = -36x$ . Its time period is:

- (A)  $2\pi/6$
- (B)  $\pi/3$
- (C)  $1/3$
- (D)  $\pi/6$

x. The wavelength and power of incident light are  $4000 \text{ \AA}$  and  $0.1 \text{ W}$  respectively. The number of incident photons per second is:

- (A)  $2.011 \times 10^{17}$
- (B)  $3.011 \times 10^{17}$
- (C)  $4.011 \times 10^{17}$
- (D)  $5.011 \times 10^{17}$

### SECTION – (A Very Short Answers)

**Q2. Answer the following: (10M)**

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- i. What is the velocity of a particle performing SHM at its extreme position?
- ii. What happens to a ferromagnetic substance when heated above Curie temperature?
- iii. What is the coefficient of transmission for athermanous materials?
- iv. On which property does dark current in a photodiode depend?
- v. Define mutual inductance.
- vi. What is magnetization?
- vii. In a hydraulic lift, area of pistons are  $A_1$  and  $A_2$  ( $A_2 > A_1$ ). If force  $F$  is applied on piston 1, what is force on piston 2?
- viii. If pressure of a gas in closed vessel increases by 0.4% when heated by  $1^\circ\text{C}$ , find initial temperature.
- ix. A metal ring of radius 20 cm is rotating about transverse axis. What is its radius of gyration?
- x. If intensity of unpolarized light reduces to half after polarizer, what is Malus' law result?

### SECTION – B (Attempt any EIGHT)

- Q3. Obtain expression for moment of inertia of a uniform disc of mass  $M$  and radius  $R$  about an axis through its centre and perpendicular to its plane.
- Q4. Effect of impurities on surface tension of liquid.
- Q5. Explain thermodynamics of adiabatic process for diatomic gas.
- Q6. Draw neat labelled diagram of kinetic energy and displacement in SHM.
- Q7. Define cohesive force, range of molecular forces, and sphere of influence.
- Q8. Distinguish between diamagnetic and paramagnetic materials.
- Q9. State Faraday's laws of electromagnetic induction.
- Q10. In a purely resistive AC circuit  $V = 140 \sin(314t)$ . If RMS current is 1.98 A, find resistance.
- Q11. Equation of motion of SHM is  $x = -36x$ . Calculate time period.
- Q12. Two tuning forks of frequencies 160 Hz and 170 Hz produce beats. Find beat frequency and wavelengths (speed of sound = 326.4 m/s).
- Q13. Show condition for first diffraction minimum for single-slit at  $49.5^\circ$ .
- Q14. Potential gradient of potentiometer wire is 0.5 V/m. If emf of cell balances at 216 cm, find emf.

### SECTION – C (Attempt any EIGHT) (24M)

- Q15. Obtain expression for work done in an isothermal process. Draw P–V diagram.
- Q16. Define: cohesive force, angle of contact, capillarity.
- Q17. Show only odd harmonics are present in vibrations of air column closed at one end.
- Q18. Expression for electric intensity outside uniformly charged spherical shell.

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- Q19. With neat circuit diagram explain use of meter bridge to determine unknown resistance.
- Q20. Derive expression for magnetic field at center of circular arc carrying current  $I$ .
- Q21. Derive radioactive decay law and state SI unit of activity.
- Q22. Find shortest and longest wavelengths of Lyman series of hydrogen.
- Q23. Show ratio of de-Broglie wavelengths of electron and proton for same kinetic energy.
- Q24. A 120 W filament lamp loses all energy by radiation. Emissivity = 0.6, area =  $3.53 \times 10^{-3} \text{ m}^2$ . Calculate surface temperature.
- Q25. A parallel plate capacitor is maintained at 400 V. Area =  $40 \text{ cm}^2$ , separation = 40 mm. Calculate displacement current.
- Q26. A Carnot engine receives 4000 J heat at 500 K and rejects 2000 J at 200 K. Find:  
(i) Work done  
(ii) Efficiency

**SECTION – D (Attempt any THREE) (12M)**

- Q27. Define isothermal process. Obtain expression for work done during such a process.
- Q28. Using phasor diagram for LCR circuit, derive expression for impedance. Define admittance.
- Q29. Obtain expression for motional emf in rotating rod in uniform magnetic field.
- Q30. Define logic gate. Explain NOT gate.
- A CE-configured transistor has DC gain 50. If collector current is 3 mA, find base current.
- Q31. State four conditions for obtaining steady interference pattern.

